PS-41 Project Brief and Work Limits

Brookside Road stormwater pump-station excavation - source record 001

1. Project Summary

PS-41 is a stormwater pump-station excavation adjacent to Brookside Road and the Mill Creek service canal. The package defines project-specific work limits, staging constraints, and required reviews for planning only.

Record ID	Item	Supplied value	Project use
BR-001	Project name	PS-41 Brookside Road Stormwater Pump Station	Source package identifier
BR-002	Work area	Station 5+040 to 5+112, north lane of Brookside Road	Traffic and utility staging limit
BR-003	Pump-station excavation	Sheet-piled box, 7.20 m by 6.40 m clear working area, design bottom elevation 96.40 m	Excavation and temporary support basis
BR-004	Final two-lane reopening milestone	2026-10-17 06:00	Target for restoration planning
BR-005	Base slab target placement	2026-10-09 10:00	Concrete sequence target
BR-006	Wet-well section target set date	2026-10-13 08:00	Installation sequence target
BR-007	Submittal boundary	Temporary works package is for constructability review only	No approval or certification language permitted
BR-008	Coordinate basis	Project grid PS41-Northing/Easting, no public survey control supplied	Use only supplied project coordinates

2. Work Limits and Access Constraints

Constraint ID	Area	Project-specific requirement	Applies to	Planning consequence
WL-001	Work hours	Normal work window is 07:00 to 18:00 Monday through Friday. Night and weekend work are not listed as permitted.	All stages	Any night or weekend activity requires a corrected project record.
WL-002	Peak traffic	At least one traffic lane must remain open from 07:00 to 09:00 and from 16:00 to 18:00.	Traffic staging	Lane-retention checks govern excavation and deliveries.
WL-003	Full road closure	Full road closure is not permitted on Brookside Road.	Traffic staging and crane picks	Any full-closure row conflicts with the brief.
WL-004	Minimum traffic lane width	Any retained traffic lane must be at least 3.25 m clear.	Traffic staging	Physical layout must demonstrate the clear lane.
WL-005	Emergency access	Emergency-vehicle passage must maintain 3.50 m clear width at all times.	Traffic and pump staging	Emergency access width is a hard project constraint.
WL-006	Driveway access	Residential driveway access must be restored, bridged, or escorted at least once every 4 hours.	Frontage access	Long single-lane operations need access handling.
WL-007	Pedestrian route	Temporary pedestrian route must be signed and at least 1.50 m clear.	Sidewalk detour	Traffic and permit data must agree on route width.
WL-008	Open excavation exposure	Continuous open excavation at the live-lane edge is limited to 18 hours.	Excavation and shoring	Excavation sequence needs an exposure-duration check.

3. Temporary Works and Excavation Controls

Control ID	Control area	Required input or limit	Required timing	Planning note
SH-001	Temporary works design package	Temporary works design reference and reviewer signoff must be recorded before excavation.	Before excavation below pavement subbase	Do not infer temporary works approval from the layout file.
SH-002	First internal brace	First brace must be installed before excavation exceeds 1.50 m below existing grade.	Before deeper excavation	Compare against planned excavation depth at first brace installation.
SH-003	Second internal brace	Second brace must be installed before excavation exceeds 3.60 m below existing grade.	Before lower-stage excavation	Compare against planned depth at second brace installation.
SH-004	Sheet-pile toe embedment	Minimum toe embedment is 3.00 m below design excavation bottom.	Before excavation release	Layout toe levels must demonstrate embedment.
SH-005	Base inspection	Formation acceptance is required before blinding concrete or base slab reinforcement.	After final trim	Hold point needed before concrete sequence.
SH-006	Vibration trigger	Peak particle velocity trigger is 5.0 mm/s at nearest utility or building monitor.	During sheet-pile driving	Baseline and readings required for trigger review.

4. Dewatering, Groundwater, and Discharge Controls

Control ID	Control area	Project-specific limit	Applies to	Planning consequence
DW-001	Dewatering start	Dewatering must be operational 12 hours before excavation below elevation 100.60 m.	Excavation below groundwater	Dewatering setup is a predecessor to lower excavation.
DW-002	Minimum duty capacity	Minimum operating duty capacity is 55 L/s, and standby units are excluded from duty capacity.	Dewatering plant	Duty pump capacity must be checked against calculated requirement.
DW-003	Discharge point	Dewatering discharge is allowed only to MH-D7 with sediment treatment installed.	Discharge operation	Alternate discharge routes are not allowed by this brief.
DW-004	Turbidity limit	Discharge turbidity must not exceed 25 NTU above background.	Discharge monitoring	Monitoring records above this trigger require action.
DW-005	Drawdown limit	Drawdown at adjacent building monitoring wells must not exceed 1.20 m from baseline.	Groundwater monitoring	Monitoring logs above this trigger suspend deeper excavation pending review.
DW-006	Fuel setback	Fuel storage, refueling, and pump generator fuel tanks must be at least 15 m from the canal edge and drainage inlet.	Pump setup	Pump staging must satisfy environmental setbacks.
DW-007	Secondary containment	Fuel containers must have secondary containment of at least 110 percent of stored volume.	Pump setup	Containment volume must be verified before operation.

5. Concrete, Wet-Well, and Restoration Controls

Control ID	Concrete or restoration item	Project-specific requirement	Planning note
CN-001	Base slab pour	Base slab concrete must be placed as a continuous pour with no unplanned cold joint.	Delivery and discharge timing must support a single continuous placement.
CN-002	Wet-well installation release	Wet-well sections must not be set until 72 hours after base slab concrete placement and cure verification.	Wet-well setting date must follow the cure release.
CN-003	Backfill release	Backfill against wet-well annulus grout must not start until 48 hours after grout placement and inspection signoff.	Backfill sequence is cure-dependent.
CN-004	Concrete certification	Concrete ticket and certificate ID must be recorded before base slab placement.	Missing certificate details remain open.

Control ID	Concrete or restoration item	Project-specific requirement	Planning note
CN-005	Asphalt reinstatement	Permanent asphalt requires dry surface and air temperature at least 7 deg C.	Weather input must be provided before paving.
CN-006	Temporary surface	Cold patch is not accepted as final reinstatement.	Final reopening logic must use permanent surface restoration.